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Magnetic Polarity Stratigraphy of Pliocene-Pleistocene Terrestrial Deposits and Vertebrate Faunas, San Pedro Valley, Arizona

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ABSTRACT

Horizontally stratified magnetic zones are observed throughout the Pliocene-Pleistocene valley fill of the upper San Pedro Valley, Arizona. Within one detailed 150-m stratigraphic section (Curtis Ranch), twelve superposed magnetic polarity zones are identified spanning the time from the upper Gilbert Epoch through the Brunhes Epoch. These magnetic polarity zones extend over a 64-km stretch of the valley and serve as convenient chronostratigraphic units. Based on the position of mammalian fossils within the chronostratigraphic zones, we have defined four mammal datum planes for the upper San Pedro Valley, including the first appearance of *Lepus* sp. at 1.9 m.y. *Lepus* sp. marks the first occurrence of a definitive Irvingtonian land mammal in the San Pedro Valley and thus places the local boundary of the Irvingtonian Land Mammal Age shortly below the Olduvai event. This position equates approximately with the Pliocene-Pleistocene boundary of the pelagic marine record. *Key words: geochronology, stratigraphic paleontology, mammaliam paleontology, paleomagnetism.*

INTRODUCTION

Cenozoic chronology was founded on biostratigraphy, with a terrestrial system based on specific events of mammalian evolution, and a comparable marine system based on molluscan or foraminiferal faunal events. The correlation of continental and marine strata is difficult because these mutually exclusive environments of deposition preclude the frequent interdigitation of diagnostic faunas and floras. As a consequence, resolution within and between the continental and marine records has reached its practical limits with conventional biostratigraphic correlation techniques. Within the pelagic marine record, however, correlations have been refined to a remarkable extent in recent years. This has been possible because of the sudden development of magnetic polarity stratigraphy as an accurate absolute correlation tool (Opdyke, 1972). The success of remanent magnetic studies in defining the stratigraphy of pelagic sediments has invited its use on other sediment types, especially terrestrial sedimentary deposits. If, indeed, mammal-bearing terrestrial deposits are generally amenable to the techniques of magnetic polarity stratigraphy, then intercontinental scale, absolute-time correlations with the marine record are possible. This development would then make possible the combination of all the relevant marine and continental data into a comprehensive chronology. The present study is one step to this end, specifically, the magnetic polarity stratigraphy of a well-known, mammal-bearing deposit in North America.

Magnetic polarity stratigraphy has been extensively used on sequences of volcanic rocks (Roche, 1950, 1951; Hospers, 1954; Rutten and Wensink, 1960; and many others), but its use in terrestrial sedimentary deposits has been rather restricted. Russian workers were among the first to use the technique in the Pliocene and Quaternary sections of western Turkmenia (Khramov, 1958). Other noteworthy studies on terrestrial sediments include those by Griffiths (1955), Bucha and others (1969), van Montfrans and Hospers (1969), van Montfrans (1971), and Wensink (1972). This study serves to extend this type of work to the valley-fill sediment of western North America.

GEOLOGIC SETTING

The San Pedro Valley near Benson, Arizona, consists of a typical basin in the Basin and Range province with a valley fill of Miocene-Pleistocene age. The deposits of the upper San Pedro Valley were studied by Gray (1967), who defined and named the St. David Formation. The St. David comprises a diversified valley fill more than 180 m thick consisting of lacustrine, swamp, and fluvial deposits in a complex mixture of lithologies including gravel, silt, clay marl, and paleosol deposits. Gray (1967) divided the St. David Formation into three members. The lowest member consists of unfossiliferous red clay and silt, sometimes gypsiferous; the middle member contains primarily lacustrine and paludal strata and bears most of the fossil record. The upper member marks a distinct change in sedimentary regimen, with a conspicuous absence of marl and clay and the presence of sand, gravel, and thick paleosol deposits. The St. David Formation was subjected to mild local deformation and a rather extensive phase of erosion following its deposi-

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tion. This erosion has incised a considerable relief onto the former landforms. The episode of erosion was accompanied by a pervasive gravel veneer deposited over the St. David Formation. The term "granite wash" is informally used to depict this gravel layer. The upper surface of the granite wash layer forms the "Whetstone Surface" of Bryan (1926; Gilluly, 1956). Lastly, the Whetstone Surface has been subjected to local degradation and aggradation, leaving irregular lenticular bodies of marl, gravel, and soil over the Whetstone Surface (Haynes, 1967).

Intermittently, ash beds are found within the St. David Formation. These ash deposits represent several volcanic episodes that have been described by Scarborough (1974). The surficial deposits described by Haynes (1967) fall largely in the C^{14} dating range.

The present San Pedro River courses through the valley from south to north and is cutting an erosion scarp into the older Whetstone Surface. This scarp and its associated badlands afford the best natural outcrops of the St. David Formation. It is also in this scarp-badland zone that most vertebrate fossils are found.

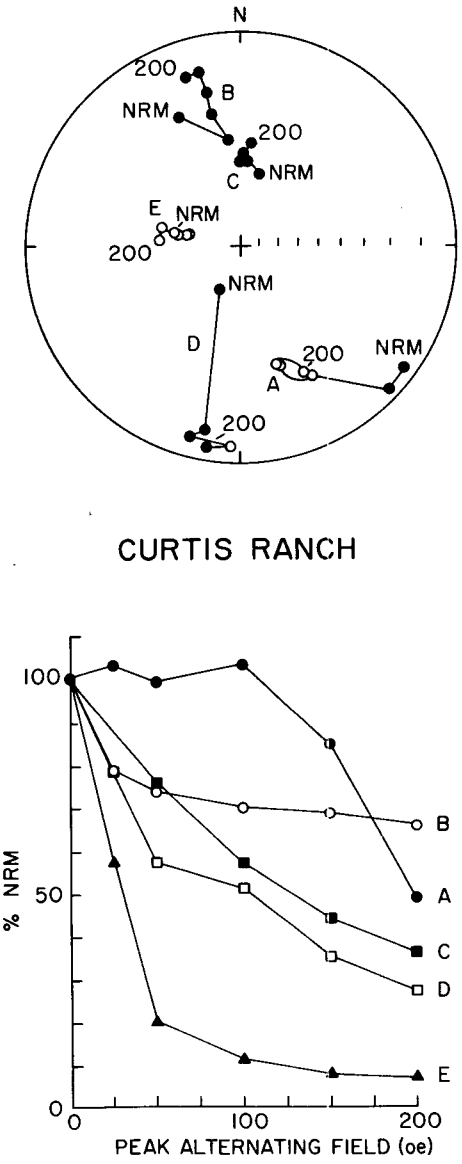


Figure 1. Progressive change in direction of magnetization with increasing alternating fields for selected samples plotted on stereographic projection (top). Filled circles are lower hemisphere projections. Alternating field demagnetization curves for same samples (bottom).

PALEONTOLOGY

Vertebrate fossils were discovered in the San Pedro Valley in 1920 by Kirk Bryan. Fossils were subsequently collected and studied by Gidley (1922, 1926), Gazin (1942), and Lammers (1970). Surficial deposits in the San Pedro Valley were studied by Haynes (1967), who compiled the record for the late Pleistocene (Rancholabrean Land Mammal Age) fauna.

Gidley (1922) described and named the Benson and Curtis Ranch faunas in the San Pedro Valley. He recognized that these faunas are slightly different in age, even though they are nearly from the same stratigraphic level and have structural similarities. Gazin (1942) reported that the Benson fauna falls definitely in the Blancan Land Mammal Age (late Pliocene) and agreed that the Curtis Ranch fauna is younger than the Benson fauna. Savage (1951) proposed a new land mammal age, Irvingtonian, for post-Blancan and pre-Rancholabrean faunas. Since 1951 the Curtis Ranch fauna has generally been considered to fall within the Irvingtonian Land Mammal Age.

Thirteen fossil sites in the San Pedro Valley have been selected here for faunal analysis. Fifty-four fossil mammal taxa (listed in Table 1) represent the known faunal record for the 13 sites. These sites were also sampled and analyzed for their paleomagnetism and their position on the magnetic time scale (Cox and others, 1964). The Post Ranch site corresponds to the Benson fauna of Gidley and Gazin. The Gidley site and Glyptotherium site correspond to the Curtis Ranch fauna of Gidley (1922).

METHODS AND MATERIALS

Paleomagnetic samples were taken from 160 sites along 40 mi (64 km) of the upper San Pedro Valley. However, a major effort was concentrated in the area where the Curtis Ranch fauna occurs. Here a long, rather continuous exposure of rocks provided us with our longest and most complete record. Our final sample suite at Curtis Ranch consisted of 81 sites, with some 240 samples, averaging about 3.3 m in vertical separation when replication is accounted for. About 150 m of stratigraphic section is represented by this sample suite. On a similar but less intense scale, all the better-known fossil localities in the upper San Pedro Valley were sampled and studied to establish their magnetic-stratigraphic position.

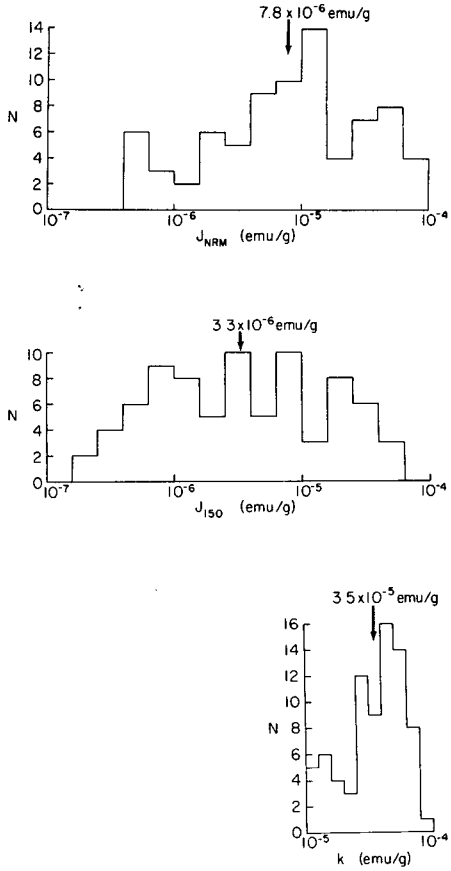


Figure 2. Histograms of natural remanent magnetization (J_{NRM}), intensity of magnetization after demagnetization in 150 oe (J_{150}), and susceptibility (k). Based on 78 specimens from the Curtis Ranch section. Geometric means appear above arrows.

Each site was sampled by taking three separately oriented specimens at three distinctly separated locations within a sedimentary unit. This sampling procedure is the minimum that will assure good site statistics and unambiguous polarities. The magnetic data reported in this study are for the most part derived from three measurements per site. In some cases, however, fewer measurements were made because of problems such as sample disintegration.

The preferred lithology for magnetic sampling is clay, which composes the bulk of samples collected. However, silt and, more rarely, sand and marl were also collected when suitable clay material could not be obtained. Specimens were oriented in the field by filing a horizontal or vertical plane on the specimen with hand rasp; a magnetic vector (measured by Brunton compass) was

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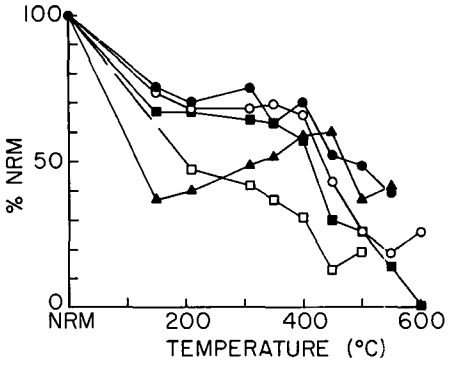


Figure 3. Thermal demagnetization curves for selected samples.

inscribed on the plane. Samples were then broken from the outcrop and sent to the laboratory. There the samples were sized and shaped to fit the magnetometer sample compartment by sawing and finally by grinding on a disk sander. The error associated with these consecutive mechanical operations is considered ±5° in both declination and inclination.

Samples were selected in the field by their stratigraphic position, lithology, and state of weathering. In deeply weathered outcrops, hand excavations were made to obtain fresher sample material. Along with collections of the magnetic samples, a detailed description and measurement of the stratigraphy was made. Elevation measurements were made by hand leveling between bench marks or reliable topographic datum points. The entire section was "walked out" by means of convenient marker beds, usually marls. In this terrane and climate, the marl units characteristically form conspicuous ledges or surfaces. Initial dip measurements were made by plane table methods. The maximum measured dip was about 12 m of rise to 1.6 km (approximately 30 minutes of arc).

MAGNETIC MEASUREMENTS

The direction and intensity of remanent magnetization was measured in a 5-cps fluxgate spinner magnetometer of the type

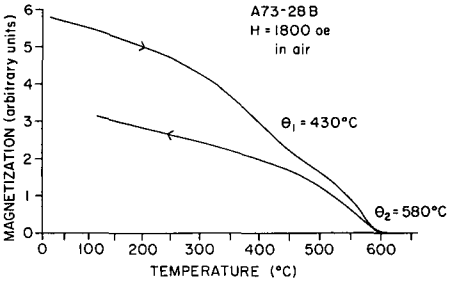


Figure 4. Thermomagnetic curve for specimen of red clay. Θ_1 and Θ_2 are Curie temperatures of different magnetic phases.

described by Foster (1966). Alternating field (a.f.) demagnetization was carried out on selected samples with equipment of the type described by McElhinny (1966). Resulting curves of five specimens along with the concomitant change of direction are shown in Figure 1. It can be seen that the sediment types used in this study possess a broad range of stabilities. However, in most of the cases shown, the median destructive field is above 100 oe, indicating that a component of considerable stability is present in the rock. The geometric mean value for the natural remanent magnetization (NRM) is 7.8×10^{-6} emu/g, ranging from 4×10^{-7} emu/g to 8×10^{-5} emu/g (Fig. 2). The fact that a range of stabilities and intensities are represented is not surprising considering the variety of lithologies used in this study. The average intensity of magnetization after demagnetization in 150 oe is 3.3×10^{-6} emu/g, which indicates that the average median destructive field is between 100 and 150 oe. This denotes relatively good stability. The susceptibility k was measured on a commercially available susceptibility bridge. The geometric mean value for k was 3.5×10^{-5} emu/g, which is higher than the

TABLE 1. FAUNAL LIST—SELECTED SITES IN THE SAN PEDRO VALLEY, ARIZONA

Taxa	Post Ranch	Mendevil Ranch	Bonanza	Honey's Hummock	McRae Wash	Horsey green bed	Wolf Ranch	Cal Tech	Calif. Wash	Johnson Pocket	Gidley level	Glyptotherium	Prospect
Sorex sp. (shrew)							X		X				
Talpidae (mole)											X		
Simonyateris stocki (bat)							X	X					
Eptesicus sp. (brown bat)					X								
Glyptotherium arizonae (glyptothere)											X	X	
Leporidae (rabbit)		X			X								
Notolagus sp. (rabbit)							X						
Aluralagus sp. (rabbit)	X	X											
Sylvilagus sp. (rabbit)									X		X		
Nekrolagus sp. (rabbit)							X						
Lepus sp. (rabbit)											X		
Spermophilus bensoni (squirrel)	X				X		X		X				
Spermophilus cochisei (squirrel)											X	X	X
Castor sp. (beaver)											X		
Geomys minor (gopher)	X	X							X				
Geomys pereimilis (gopher)							X		X		X	X	
Cratogeomys bensoni (gopher)	X	X			X								
Perognathus sp. (pocket mouse)	X	X		X	X		X		X		X		
Prodipodomys minor (kangaroo rat)	X	X		X	X		X						
Prodipodomys idahoensis (kangaroo rat)							X		X				
Dipodomys gidleyi (kangaroo rat)											X	X	
Bensonomys arizonae (small mouse)	X	X			X		X		X		X		
Peromyscus sp. (deer mouse)							X				X	X	
Baiomys minimus (pigmy mouse)	X	X			X		X						
Baiomys brachygnathus (pigmy mouse)							X					X	
Onychomys bensoni (grasshopper mouse)		X					X						
Onychomys pedroensis (grasshopper mouse)											X		
Sigmodon medius (cotton rat)	X	X		X	X		X						
Sigmodon minor (cotton rat)									X		X	X	
Sigmodon curtisi (cotton rat)							X		X		X	X	
Neotoma fossilis (pack rat)	X	X											
Neotoma sp. (pack rat)					X		X				X		
Synaptomys (Metaxomys) sp. (lemming)									X				
Pliophenacomys sp. (pine vole)		X											
Ondatra idahoensis (muskrat)									X		X		
Coendou sp. (tree porcupine)							X						
Neochoerus sp. (capybara)								X					
Canis edwardii (coyote)											X		
Canis sp. (dog)		X			X					X			
cf. *Borophagus* sp. (dog)	X												
Spilogale pedroensis (spotted skunk)											X		
Mustelidae (weasel)	X										X		
Felis sp. (cat)	X												
Panthera sp. (jaguar)											X		
cf. *Stegomastodon* sp. (gomphothere)								X			X		
cf. *Cuvieronius* sp. (gomphothere)	X	X					X		X				
Nannippus cf. *phlegon* (three-toed horse)	X	X	X	X	X	X	X						
Equus sp. (horse)	X	X		X	X	X	X	X	X		X	X	X
Platygonus sp. (peccary)	X	X			X			X		X	X		
Tanupolama sp. (llama)		X									X	X	
Camelops sp. (camel)	X	X			X		X		X		X	X	
Odocoileus sp. (deer)					X			X					
Capromeryx gidleyi (pronghorn)											X		
Antilocapridae (pronghorn)	X	X			X								

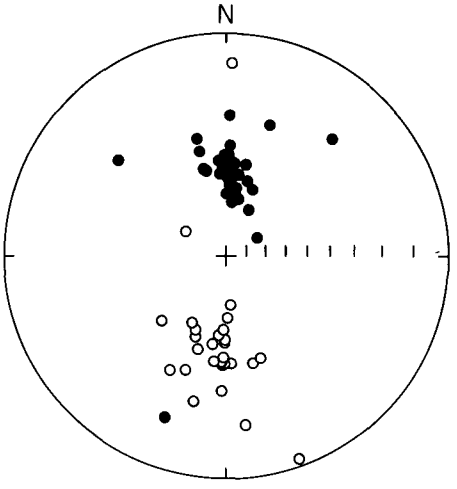


Figure 5. Site mean directions of magnetization for 57 Curtis Ranch sites after alternating field demagnetization plotted on stereographic projection. Filled circles from normally magnetized sites (lower hemisphere projections), open circles from reversely magnetized sites.

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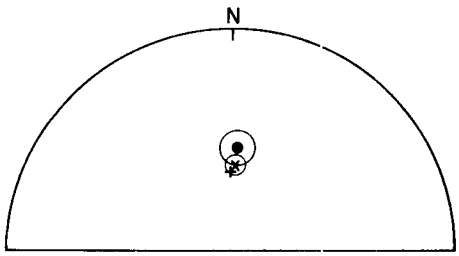


Figure 6. Mean directions of normally and reversely magnetized sites from Curtis Ranch with circles of 95 percent confidence plotted on lower hemisphere of stereographic projection.

mean intensity of magnetization. In Figure 1, the a.f. demagnetization curves shown represent sediment samples with the following lithologies: calcareous red claystone (C and D), greenish silty clay (B and E), and brown silty clay (A). Blanket demagnetization was performed on all sites in peak fields of 150 oe. This field intensity was found to be sufficient to remove most low-coercivity viscous components, which are present in almost all of the samples. This led to good groupings within most of the sites. However, some sites possessed a more stable secondary component, which required treatment in higher fields. These sites often possessed strung distributions between the present normal field and past reversed directions of the Earth's field (Sample D, Fig. 1). Although the mean directions from these sites clearly are not an adequate measure of the magnetic field at the time of deposition, usually the correct polarity of the original field is not in doubt, so that useful information is obtained from these sites even if a significant secondary component remains after a.f. demagnetization. At some other sites, particularly those that are initially weakly magnetized, the cleaning field should be less than 150 oe. However, to determine the ideal washing field for each site would require a.f. demagnetization curves on each site. This is impractical in studies such as the present one, in which the primary purpose of the study is to determine magnetic polarity.

Thermal demagnetization studies were also performed on six pilot specimens representing various lithologies (Fig. 3) to determine whether thermal demagnetization is more effective than a.f. demagnetization. No significant changes in direction occurred within these specimens up to the Curie temperature of magnetite. Because a.f. demagnetization proved to be the more efficient method of removing unwanted secondary components, this was the technique employed routinely on all sites.

To understand better the means by which these sediments become magnetized, it is necessary to identify the mineral responsible for the NRM. To do this a magnetic separate was prepared using the method described in Løvlie and others (1971). Strong field thermomagnetic curves were obtained for separates from five representative sites in the Curtis Ranch section. All heating and cooling was performed in air. The instrument employed is a vertical motion instrument utilizing an electrobalance to measure the change in magnetic intensity. The change in magnetic moment against temperature is continuously recorded on an *X/Y* plotter. The saturating field is produced by a 4-in. electromagnet.

The thermomagnetic curves are irreversible and are similar for all sites. A representative curve is shown in Figure 4. The magnetic material contained two magnetic phases, one with a Curie temperature of about 400°C and the second (magnetite) with a Curie temperature of 580°C. The phase with the Curie temperature at 400°C cannot be identified with certainty but is probably a titanomagnetite or titanomaghemite. This magnetic phase may be responsible for the consistent drop in magnetic intensity at 400°C noted during the thermal demagnetization experiments (Fig. 3). Because the principal magnetic mineral within these sediment samples is clearly magnetite and the layers are not substantially weathered, their remanence is most likely detrital remanent magnetization. It is therefore most probable that the remanence dates from the time of the deposition of the sediment. The particles of detrital magnetite are most likely derived from the mountains proximal to the San Pedro basin.

Figure 5 shows the site mean directions of magnetization for 57 Curtis Ranch sites where the best estimate of the precision is greater than 10 (Fisher, 1953). The site mean direction for the normally magnetized sites is declination 3.3°, inclination +48.1°; and for the reversely magnetized sites, it is declination 186.4°, inclination −39.6°. The circles of 95 percent confidence (Fisher, 1953) for the normally and reversely magnetized sites are 5.1° and 7.0°, respectively. Figure 6 shows that there is no significant difference between the normal and reversed

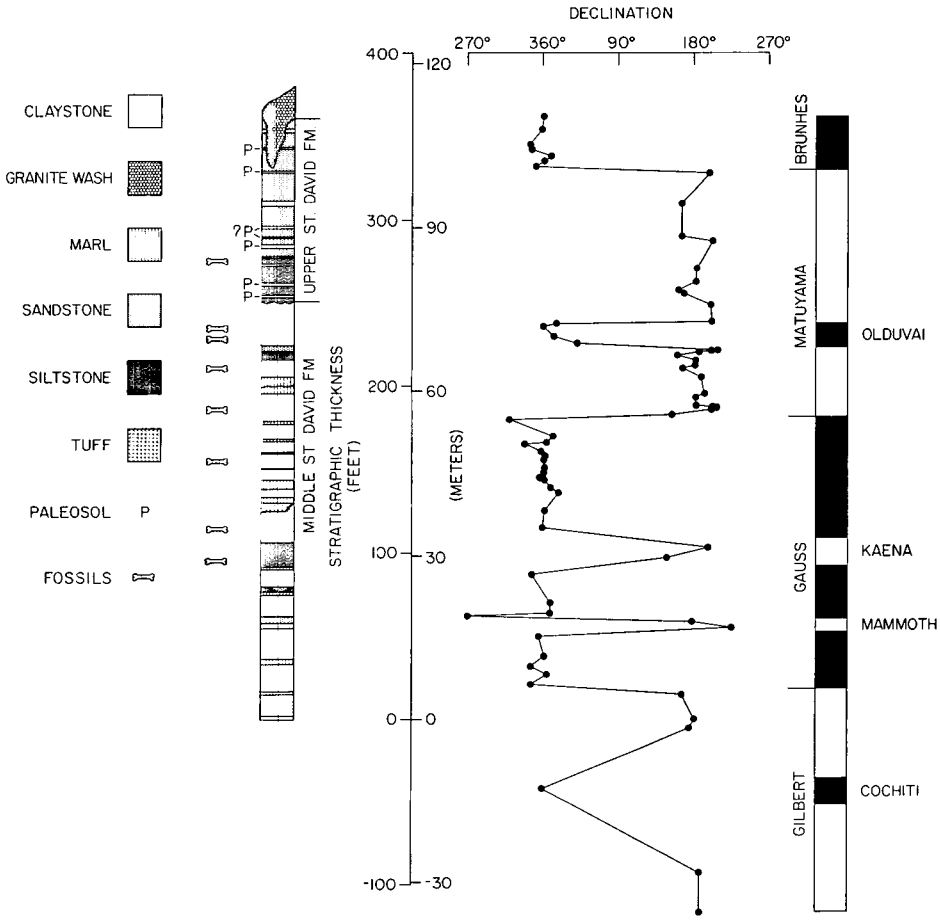


Figure 7. Variation of magnetic declination with stratigraphic level at Curtis Ranch. In magnetic time-scale column, black sections are normally magnetized and white are reversely magnetized. Eight fossil levels named in order of increasing age: Prospect, Glyptotherium (of Gidley), Gidley level, Johnson Pocket, Cal Tech, Horsey green bed, Honey's Hummock, and Bonanza.

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sites. The circle of 95 percent confidence from the normally magnetized sites includes the dipole field at the sites. The dipole field does, however, fall outside the circle of confidence of the reversely magnetized sites. This may be due to incomplete removal of secondary components of magnetization in the direction of the Earth's present field. The fact that the mean normal and reversed directions are antiparallel leads us to believe that the sedimentary deposits are accurately recording the changes of the Earth's magnetic field.

MAGNETIC POLARITY ZONES WITHIN THE ST. DAVID FORMATION

The variation of magnetic declination with stratigraphic level at Curtis Ranch is shown in Figure 7. It can be seen that the Granite Wash unit and the uppermost St. David Formation are normally magnetized, and below this level 11 complete magnetic transitions are recorded. Our interpretation of these magnetic polarity zones in terms of the time scale of Cox and others (1964) is given on the right-hand side of the diagram. We believe that the section spans the time from the upper Gilbert Epoch into the Brunhes Epoch. The mammal-bearing sequence ranges from the upper Gauss Epoch to the lower Matuyama Epoch, which places it, in terms of marine stratigraphy at least, within upper Pliocene and lowest Pleistocene time. This time interval includes at least a part of the Blancan Land Mammal Age (Savage and Curtis, 1970). The lower part of the reversed magnetic polarity zone associated with the middle and upper member of the St. David Formation can be correlated to a reversely magnetized volcanic ash. This ash, which is within the middle member of the St. David Formation, has been dated by Scarborough (1974) at 2.5 ± 0.4 m.y.[1] This radiometric age determination favors the correlation of the associated reversed-polarity zone with the Matuyama Epoch (0.7 to 2.4 m.y.) rather than the Gilbert Epoch (3.3 to 5.1 m.y.).

If our identification of the magnetic polarity zones is correct, the 11 magnetic reversals recorded at Curtis Ranch represent time planes of known absolute age (Dalrymple, 1972). The most uncertain time plane in this regard is the uppermost one, which is most likely the base of the Brunhes event. Because of the abundant conglomerate and paleosol deposits in this interval, our sampling density was not close enough to resolve events as short as the Jaramillo Epoch. Therefore, it seems most probable that the indicated transition is the Brunhes-Matuyama boundary and that the Jaramillo event is missing in this stratigraphic section (Fig. 7). The location of these absolute time planes within the section also provides a means to date the entire section. That is, the age of any stratigraphic horizon in the section may be reasonably estimated by an interpolation between the time planes that bound it. This assumes that the sedimentation of the St. David Formation, although decidedly episodic in character, sampled the Earth's magnetic field with sufficient time frequency to accurately record its history of polarity changes. By this reasoning, then, we can say that the fossiliferous strata in the Curtis Ranch section (Fig. 7) extend discontinuously over a 1-m.y. period, from just above the Kaena event (~2.7 m.y.) to just above the Olduvai event (~1.7 m.y.).

The vertical succession of magnetic polarity zones at Curtis Ranch is repeated also on the opposite side of the San Pedro Valley, as shown in Figure 8. In fact these same magnetic polarity zones may be traced in plan view over a 64-km reach of the San Pedro Valley (Fig. 9), in a manner reminiscent of the "magnetic stripes" of the sea floor. Our best estimate of the elevation of the principal magnetic transitions along this length of the San Pedro Valley is shown in Figure 10. If our interpretation of the magnetic polarity stratigraphy is correct, then Figure 10 depicts the absolute chronostratigraphic zonation of the St. David Formation. It may be used, therefore, to infer the absolute ages for various stratigraphic intervals and fossil sites along this stretch of the San Pedro Valley.

STRATIGRAPHIC PALEONTOLOGY

Figure 11 shows the superposition of 13 selected fossil sites within the upper San Pedro Valley. These sites are ordered according to their observed lithostratigraphic and magnetic-stratigraphic position. Absolute ages of the sites can be interpolated from our magnetic-stratigraphic column (Fig. 11), using the age assignments of Opdyke (1972).

We recognize four important temporal datum planes in the fossil sequence of the San Pedro Valley. These datum planes specify the time at which minor but significant changes occurred in the total vertebrate fossil record. They are similar in concept to the datum planes used in Cenozoic planktonic and foraminiferal zonation (Hornibrook and Edwards, 1971),

[1] RS–35a–68 of the University of Arizona Laboratory of Isotope Geochemistry. The sample is a pure, crystal-poor, air-fall vitric ash, brown, 35 cm thick. Glass shards were dated. Analytical data: K = 2.55 percent, Ar^{40}_{rad} = 12.0 × 10^{-12} m/g, atmospheric Ar = 90 percent. A different ash bed immediately underlying the Post Ranch faunal horizon (see Fig. 11) has also been dated at 3.1 ± 0.7 m.y. by the fission-track method on zircon (R.A.K. Tahirkheli, 1974, personal commun.).

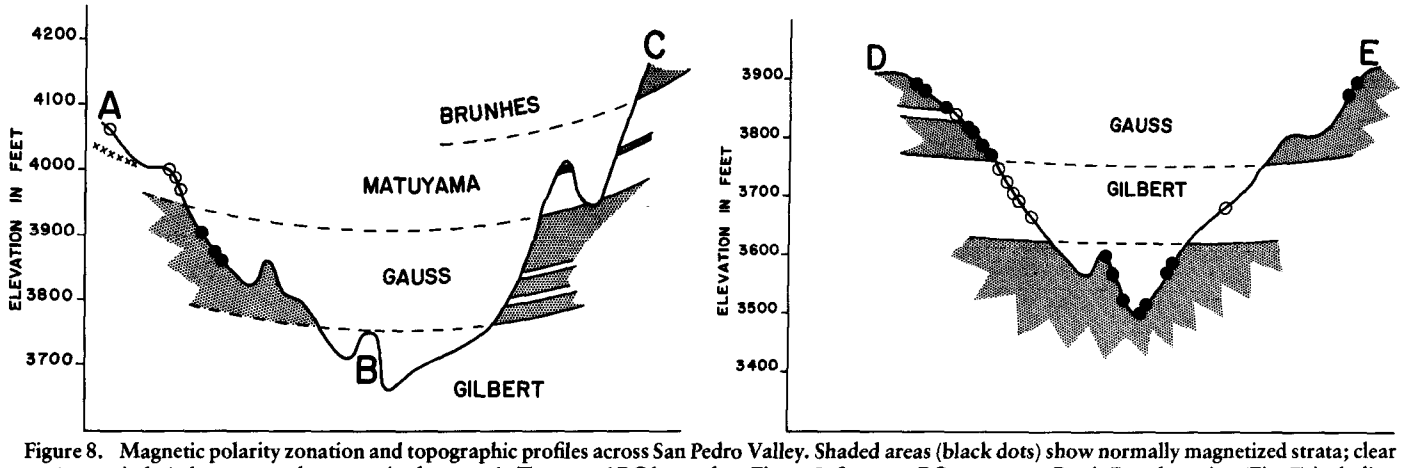


Figure 8. Magnetic polarity zonation and topographic profiles across San Pedro Valley. Shaded areas (black dots) show normally magnetized strata; clear areas (open circles) show reversely magnetized strata. A. Transect ABC located on Figure 9. Segment BC represents Curtis Ranch section (Fig. 7) including 81 sites, which are not shown. Segment AB includes California Wash faunal site, located just above Matuyama-Gauss boundary and below volcanic ash zone whose stratigraphic position is indicated by checkered pattern. This ash has been dated at 2.5 ± 0.4 m.y. (Scarborough, 1974). B. Transect DE located in Figure 9. Includes Mendevil Ranch faunal site, which is located just above Gilbert-Gauss boundary near point D.

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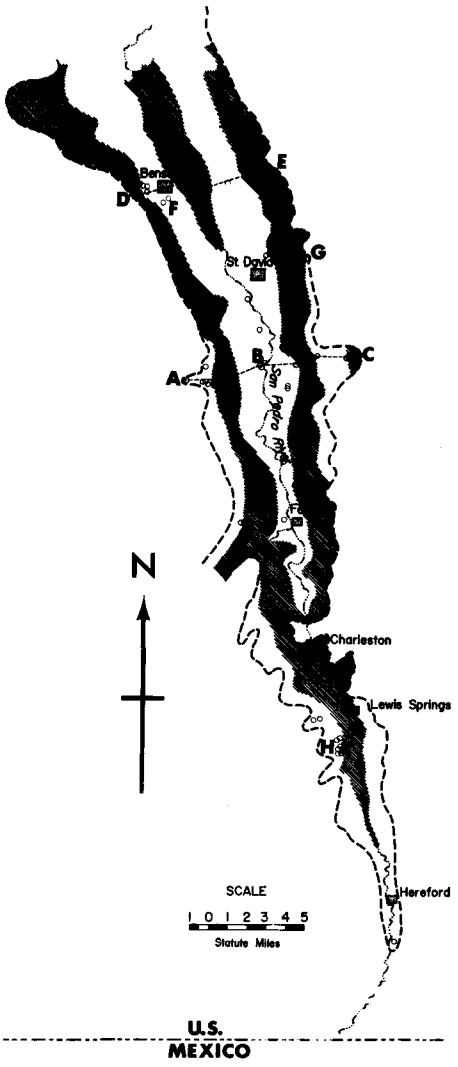


Figure 9. Plan view of magnetic polarity zones observed in St. David Formation of upper San Pedro Valley. Area of exposed St. David Formation lies generally downslope from Whetstone scarp of San Pedro River and is outlined approximately by dashed line. Patterned areas (black dots), normally magnetized strata; clear areas (open circles), reversely magnetized strata. Magnetic polarity zone distribution based on 160 sample sites, 81 of which lie along line segment BC, Curtis Ranch section (Fig. 7). Locations F, G, and H indicate positions of Post Ranch, McRae Wash, and Wolf Ranch fossil localities, respectively.

but they are more restricted in geographic range, being limited to the San Pedro Valley. Our datum planes have only two dimensions; they have no duration and are therefore distinct from biostratigraphic zones.

Datum Plane I

***Sigmodon medius* Appearance (3.1 m.y.).** This datum marks the first appearance of *Sigmodon medius* in the San Pedro Valley. *Sigmodon* is not found in deposits older than Blancan Land Mammal Age. Other early Blancan faunas that contain *Sigmodon* are the 111 Ranch fauna of Arizona, the Rexroad and Deer Park faunas of Kansas, and the Sand Draw fauna of Nebraska.

Several other taxa that occur for the first time in the San Pedro Valley deposits at the *Sigmodon medius* datum are *Aluralagus* sp., *Bensonomys arizonae, Baiomys minimus, Onychomys bensoni, Neotoma fossilis, Spermophilus bensoni,* and *Pliophenacomys* sp.

Datum Plane II

***Nannippus* Disappearance (2.4 m.y.).** The three-toed horse, *Nannippus phlegon,* occurs in many Blancan faunas of Arizona and elsewhere in North America. This datum marks the latest known occurrence of *Nannippus phlegon* in the San Pedro Valley. Probably the latest known occurrence of *Nannippus* is *N. phlegon* of the Blanco fauna, Texas.

Other taxa that are absent in the San Pedro Valley above the *Nannippus* (extinction) datum are *Aluralagus* sp., *Pliophenacomys* sp., and *Onychomys bensoni*. Mammals that appear in the San Pedro Valley at this datum are *Nekrolagus* sp., *Geomys persimilis, Baiomys brachygnathus, Sigmodon curtisi,* and *Coendou* sp.

Datum Plane III

***Ondatra idahoensis* Appearance (2.1 m.y.).** This datum marks the earliest record of the ancestral muskrat, *Ondatra idahoensis*, in the San Pedro Valley. Other early records of *Ondatra* are *Ondatra idahoensis* of the Grand View fauna in Idaho, and *Ondatra* cf. *idahoensis* of the Borchers fauna in Kansas.

Glyptotherium arizonae, *Sylvilagus* sp., *Sigmodon minor*, and *Synaptomys (Metaxomys)* sp. also appear in the San Pedro Valley deposits at the *Ondatra idahoensis* datum. *Nekrolagus* sp. and *Sigmodon medius* are not known in the San Pedro Valley above this datum.

Datum Plane IV

***Lepus* sp. Appearance (1.9 m.y.).** The earliest record of the rabbit, *Lepus* sp., in the San Pedro Valley occurs at this datum. Other early occurrences of *Lepus* are *Lepus* cf. *californicus* of the Borchers fauna, Kansas, and *Lepus* sp. of the Vallecito Creek fauna, California.

Other mammal taxa that appear in the San Pedro Valley deposits at the *Lepus* datum are *Spermophilus cochisei, Castor* sp., and *Dipodomys gidleyi.* The genera *Prodipodomys, Bensonomys, Synaptomys (Metaxomys),* and *Stegomastodon* are not recorded above this datum in the San Pedro Valley.

The *Lepus* sp. datum plane is a particularly useful and important datum for two reasons. First of all, this datum marks the appearance of a form regarded as definitive for early Irvingtonian Land Mammal Age (Hibbard and others, 1965; Downs and White, 1968). Therefore, the *Lepus* datum in the San Pedro Valley must be close to the Blancan-Irvingtonian boundary according to this definition. (In large mamal associations, Savage [1951] used the appearance of *Mammuthus* to mark the base of the Irvingtonian. These alternate concepts of the basal Irvingtonian do not necessarily refer to the same point in time.) Secondly, it is close in time to the Olduvai event, within which the Pliocene-Pleistocene boundary occurs (Berggren and others, 1967). Therefore, with the *Lepus* datum at or just below the Pliocene-Pleistocene boundary, Irvingtonian faunas can be regarded essentially as Pleistocene and Blancan faunas as Pliocene. It is noteworthy, however, that this definition of the Pliocene-Pleistocene boundary (for example, Berggren and others, 1967) does not preclude substantial terrestrial climate changes during late Pliocene time.

A number of the small mammals in the San Pedro Valley stratigraphic sequence have apparently evolved in place. The following taxa are believed to reflect ancestral-descendant relations: *Geomys minor* → *Geomys persimilis, Prodipodomys idahoensis* → *Dipodomys gidleyi, Baiomys minimus* → *Baiomys brachygnathus, Onychomys bensoni* → *Onychomys pedroensis, Sigmodon medius* → *Sigmodon minor.* A study of the rates of evolution for these taxa is beyond the scope of this paper, but the excellent time-stratigraphic control will enable such a study.

The faunal datum planes as defined above are interpolated from present magnetic evidence for the San Pedro Valley and must be updated as new information becomes available.

SUMMARY AND DISCUSSION

The study presented here demonstrates that, under some circumstances at least, meaningful measurements of depositional remanence may be obtained from Neogene terrestrial sediments and that accurate chronostratigraphic horizons and magnetic polarity transitions may be localized. However, it must be pointed out that not every outcrop of terrestrial rock is amenable to an analysis as useful as that given here. The

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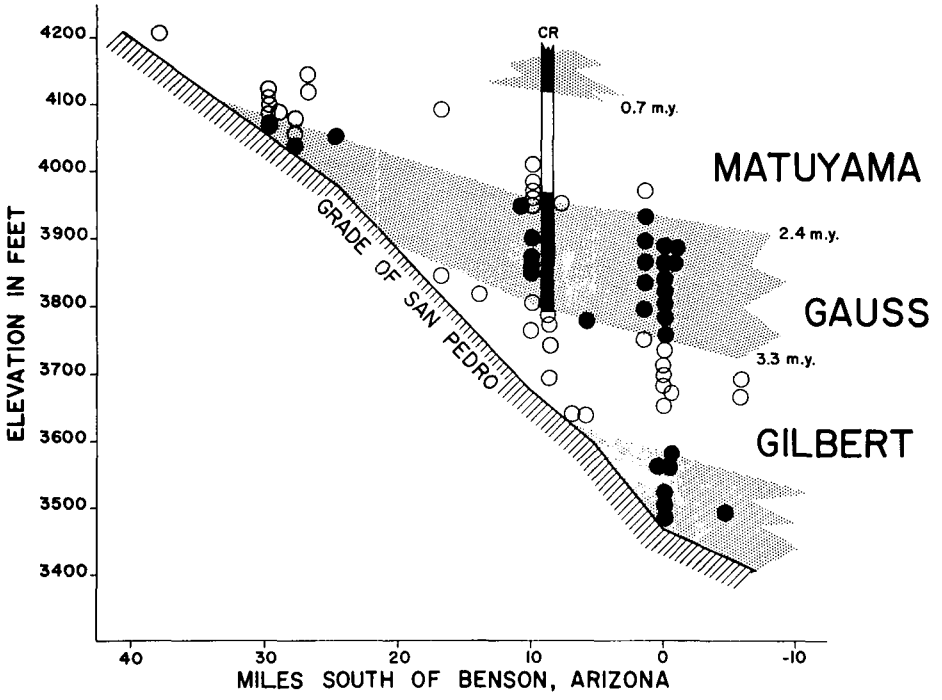


Figure 10. Vertical distribution of magnetic polarity zones within St. David Formation along upper San Pedro Valley. Shaded areas (black dots), normally magnetized strata; clear areas (open circles), reversely magnetized strata. Data base is same as described in Figure 9. Continuum of Curtis Ranch data (81 data points) represented by bar labeled CR.

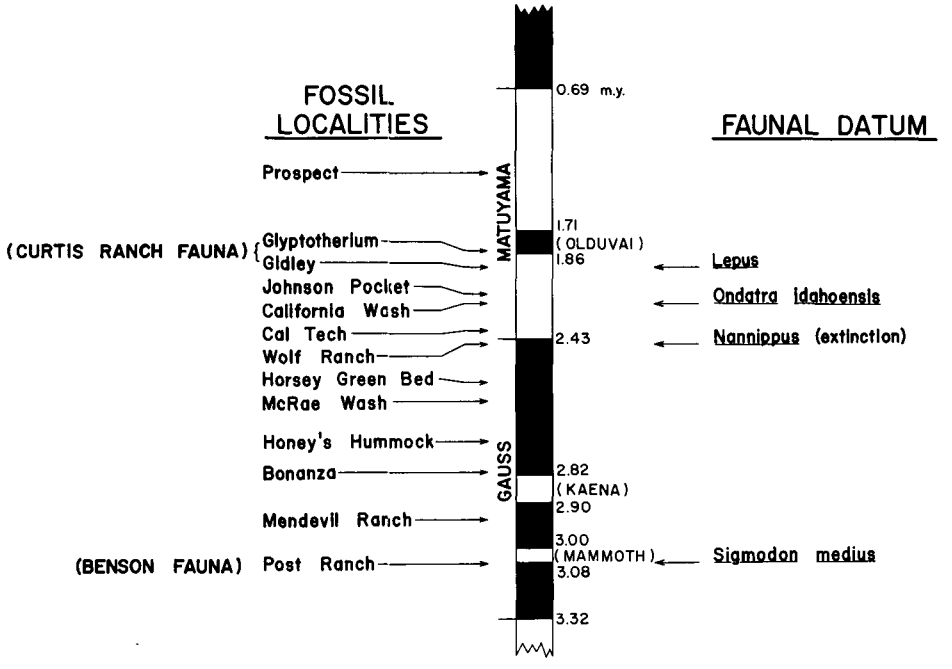


Figure 11. Temporal relations of fossil-mammal localities in upper San Pedro Valley with respect to magnetic time scale. Fossil localities include all eight of those in Curtis Ranch section (see Fig. 7) as well as five others whose position on the magnetic time scale has been directly determined (California Wash, Wolf Ranch, McRae Wash, Mendevil Ranch, and Post Ranch). Magnetic polarity zonation portrayed is that actually observed in Curtis Ranch section (Fig. 7) along with our time-scale interpretation.

Curtis Ranch section was especially favorable because of its lithology, fossil content, and long continuous depositional record.

An important limitation to the method is imposed by the quality and continuity of sediment present in a given outcrop. Fine-grained sediment, such as claystone and siltstone is the preferred lithology for reliable paleomagnetic studies. A lengthy stratigraphic column also enhances the prospects of crossing one or more magnetic transitions. At present only magnetic epochs and events from the past 5 m.y. have reliably dated boundaries.

Our experience (in addition to the San Pedro Valley, this also includes data from Meade County, Kansas; the Texas Panhandle, and the Anza-Borrego region, California) suggests that an isolated, fossiliferous outcrop can be quite easily assigned to the correct magnetic epoch, for example, Brunhes, Matuyama, and so forth, on the basis of one or two accurate magnetic polarity determinations, plus some knowledge of the associated fossils. This is explained by the fact that during the past 5.1 m.y., only 15 percent of the time is marked by magnetic polarities other than that of the principal epoch (Opdyke, 1972). That is, on a random basis, 85 percent of the time during the past 5.1 m.y. has been occupied with the major "epochs" rather than the smaller "events" within the epochs. For example, the chance of finding on a random basis a normal (+) polarity (the Jaramillo or Olduvai events) within sediments of Matuyama age are only one in eight. A Matuyama-age fossil, then, will most likely be buried in a reversely (−) magnetized rock.

Once the correct magnetic epoch has been designated for a given sediment, its age has been circumscribed on a rather broad scale. Time resolution beyond the simple designation of Epoch requires a much more extensive and intensive sampling program. At Curtis Ranch for example we found that a sampling frequency of 0.5 to 1.5 percent (2 to 3 cm every 2 to 3 m or so) was needed to find and define the pertinent magnetic events. This density of sampling may be precluded in many outcrops by an abbreviated or incomplete stratigraphic section or the absence of suitable material. A judicious selection of outcrop is thus necessary to exploit the potential of magnetic polarity stratigraphy.

The real power of magnetic polarity stratigraphy, that is, its potential for absolute time correlation on a world-wide scale, is exemplified by the faunal datum planes we have erected for the San Pedro Valley. These faunal datum planes may prove useful for comparisons and correlations with other vertebrate fossil localities. In like fashion, these faunal datum planes may be

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related directly to the faunal datum planes compiled from the marine record (Berggren, 1972). A comprehensive chronology within this time span may thus be obtained, based on magnetic criteria.

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